

Confidence Intervals for Proportions

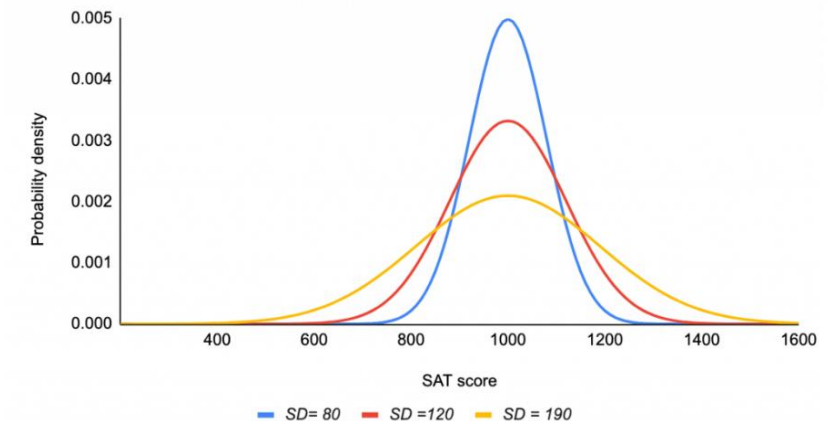
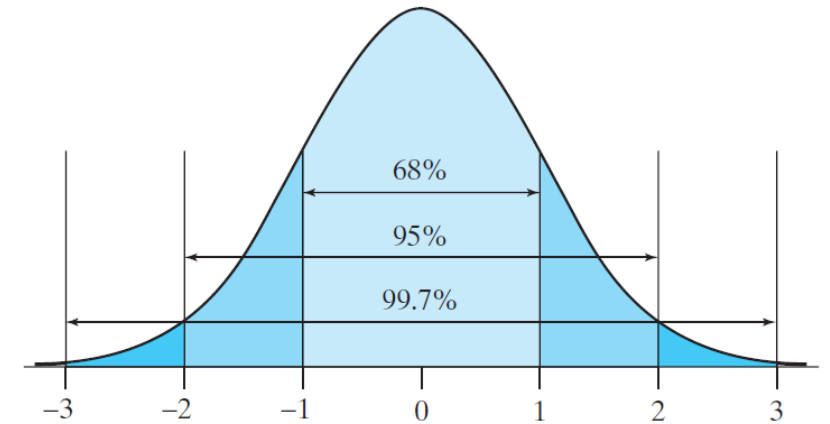
Week 3 Lecture 3

Let's put things together

- Investigation 2 – Last 4 - 5 questions
- Central Limit Theorem for Sample Proportion
- They lead us to talk about more “nicely shaped” distributions
 - Unimodal that are not too skewed and tails are not overly long or short
 - Normal Distribution

Normal Distribution

- The distribution of many real-world phenomena follow a **normal distribution**.
- Empirical Rule (**68-95-99.7% rule**): In a Normal distribution, we expect to find
 - about 68% of data within ± 1 SD of the mean.
 - about 95% of data within ± 2 SDs of the mean.
 - >99% of data within ± 3 SDs of the mean.



Review of Central Limit Theorem for Sample Proportion

REMEMBER!

- The values of statistics vary from sample to sample.
 - This phenomenon is called **sampling variability**.
 - Fortunately, if we look at the results of many samples, there is a predictable pattern to this variability.
 - This pattern is called the **sampling distribution** of the statistic.

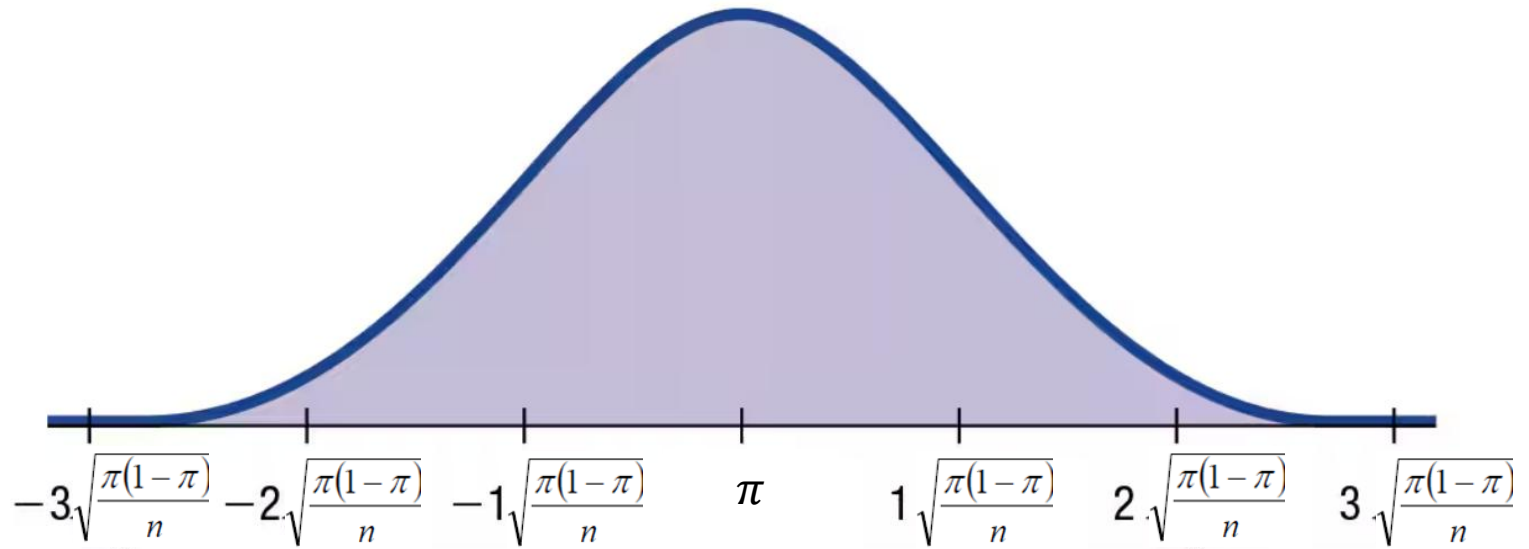
Central Limit Theorem (CLT) for Sample Proportion:

Suppose that the proportion of a population having some characteristic is denoted by π , and suppose that a random sample of size n is taken from the population. Then the sampling distribution of the sample proportion $\hat{p}$ is approximately normal with mean π and standard

deviation $\sqrt{\frac{\pi(1-\pi)}{n}}$. This approximation is generally considered to be valid as long as $n\pi \geq 10$ and $n(1-\pi) \geq 10$.

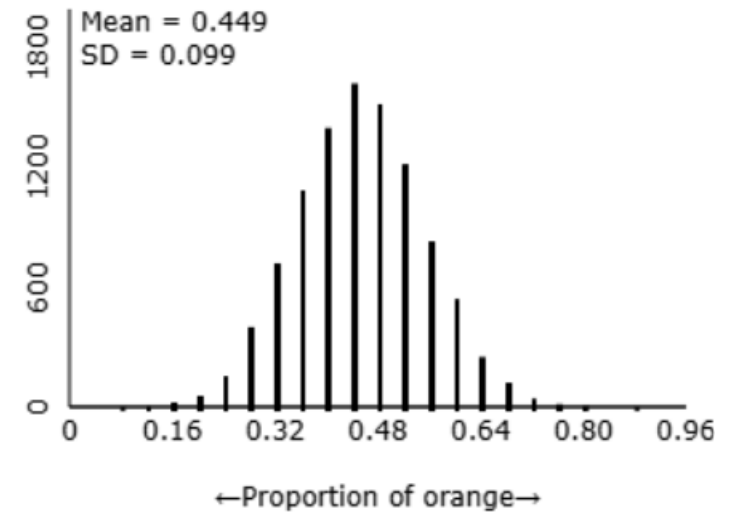
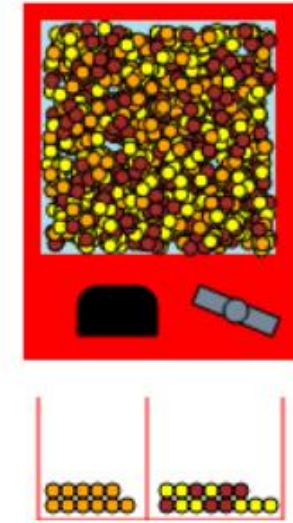
Sampling Distribution of the Sample Proportion

- Normal model centered at π with a standard deviation of $\sqrt{\frac{\pi(1-\pi)}{n}}$ is a good model for a collection of proportions found for many random samples of size n from a population with mean π .
- We've seen that the sampling distribution of proportions follows the 68–95–99.7 Rule too.



Central Limit Theorem for Sample Proportion

- Even though we can't control sampling variability, we can predict exactly how much different proportions will vary from sample to sample.
- This will enable us to make sound decisions based on a *single sample*.
 - After all, in real life, you will not take 10,000 samples, right?



Confidence Intervals for a Single Proportion

Confidence Interval for Proportions

- Much of Statistics involves making inferences about parameters based on statistics.
 - With *categorical* variables, the simplest relevant parameter/statistic is a *proportion*.
- Two primary kinds of statistical inference procedures are:
 - **Parameter estimation**, which estimate the value of a population parameter with a value or an interval of values based on a sample statistic;
 - Point Estimation
 - Interval Estimation
 - **Hypothesis Testing:** Next week...

Gallup Example

EXAMPLE: The Gallup Poll periodically asks a random sample of U.S. adults whether they think economic conditions are getting better, getting worse, or staying about the same. When Gallup polled **3559 respondents** in April 2013 (during the week ending April 21), only **1495 thought** economic conditions in the United States were **getting better—a sample proportion of 0.42.**

Gallup Example

- What can we say about consumer confidence in the entire population when the proportion that we measure keeps bouncing around from sample to sample?
- That's where the sampling distribution model can help.

Here is what we know

- What do we know about our sampling distribution model?
- It's centered at the true proportion, π , of all U.S. adults who think the economy is improving.
 - But we don't know π .
 - It probably isn't 42.0%.
 - That's the statistic (p-hat) from our sample.
- The standard deviation of the sampling distribution is $\sqrt{\frac{\pi(1-\pi)}{n}}$.
- The shape of the sampling distribution is approximately Normal, when the sample is large enough.

Is what we know useful?

- This is all fine in the model world, but we need to solve problems in the real world.
- In the real world, we don't know π .
 - If we did, we wouldn't have bothered to take a sample.
- And so, we don't know $\sqrt{\frac{\pi(1-\pi)}{n}}$ either.
 - But, we'll do the best we can and estimate it by using $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.

Is what we know useful? – Gallup Example

- We'll do the best we can and estimate it by using $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.
- That may not seem like a big deal, but it gets a special name.
- Whenever we estimate the standard deviation of a sampling distribution, we call it a **standard error (SE)**.

Gallup Example (Still!)

- Using $\hat{p}$ we find the standard error

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{(0.42)(1 - 0.42)}{3559}} = 0.008$$

What is Confidence Interval?

- Because the sampling distribution is Normal, we expect that about
 - 68% of all samples of 3559 U.S. adults would have sample proportions within 1 standard deviation of π .
 - 95% of all these samples will have proportions within $\pi \pm 2\text{SEs}$.
- But where is our sample proportion in this picture? And what value does π really have?
 - We still don't know!
 - We know that for 95% of random samples, $\hat{p}$ will be no more than 2 SEs away from π .

What is Confidence Interval?

- So here's the key to using sampling distributions.
- Let's reverse it and look at it from $\hat{p}$'s point of view.
 - If I'm $\hat{p}$, there's a 95% chance that π is no more than 2 SEs away from me.
 - If I reach out 2 SEs, or $2*0.008$ away from me on both sides, I'm 95% confident that π will be within my grasp.

Confidence Interval for Proportions

The estimated standard deviation $\sqrt{\hat{p}(1 - \hat{p})/n}$ of the sample statistic $\hat{p}$ is called **the standard error** of $\hat{p}$.

Let's calculate CI to produce a reasonable interval estimate of π

AND

Interpret its confidence interval

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{(0.42)(1 - 0.42)}{3559}} = 0.008$$

$$\hat{p} \pm 2 \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

$$0.42 \pm 2 \times \sqrt{\frac{0.42(1 - 0.42)}{3559}}$$

$$= 0.420 \pm 0.016$$

$$= (0.404, 0.436)$$

We are 95% confident that between 40.4% and 43.6% of U.S. adults thought the economy was improving

Confidence Interval for Proportions

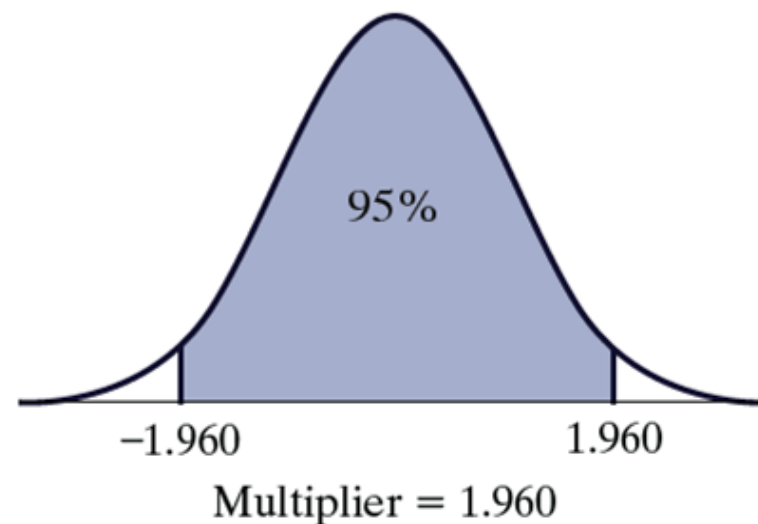
The multiplier 2 is just an approximation.

A general formula template for CI

$$\text{statistic} \pm \text{multiplier} \times (\text{SD of statistic})$$

To produce a confidence interval for the population proportion π

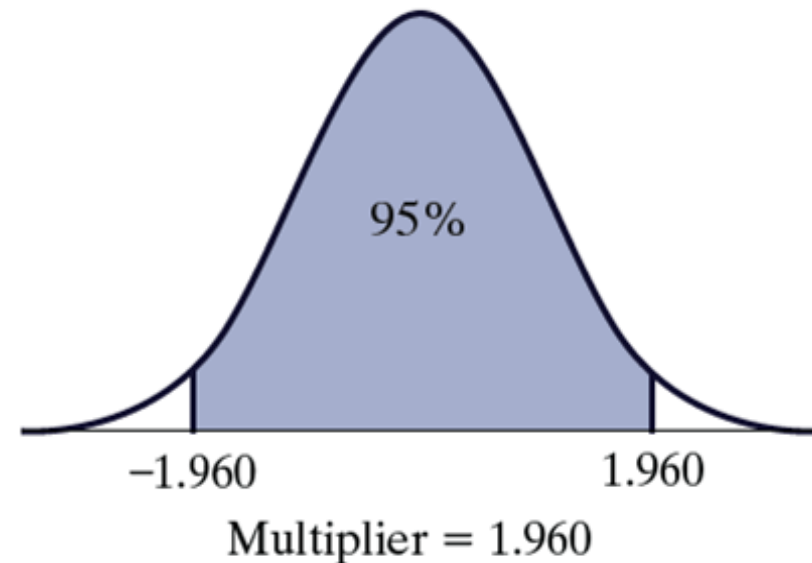
$$\hat{p} \pm \text{multiplier} \times \sqrt{\hat{p}(1 - \hat{p})/n}$$



Confidence Interval for Proportions

- This interval is called a 95% **confidence interval (CI)**.
 - The value 95% is called the **confidence level**.
 - The *multiplier* $\times SE(\hat{p})$ is called the **margin-of-error**.
 - The multiplier (sometimes call it z^*) 1.96 is called a **critical value**.

$$\hat{p} \pm \text{multiplier} \times \sqrt{\hat{p}(1 - \hat{p})/n}$$

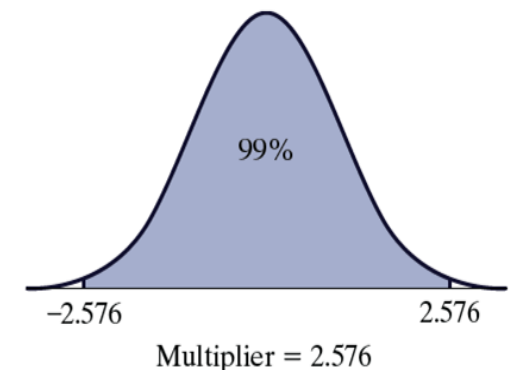
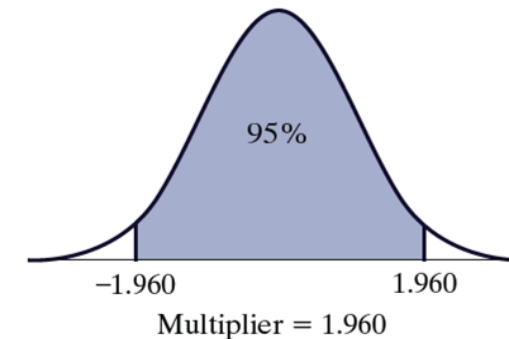
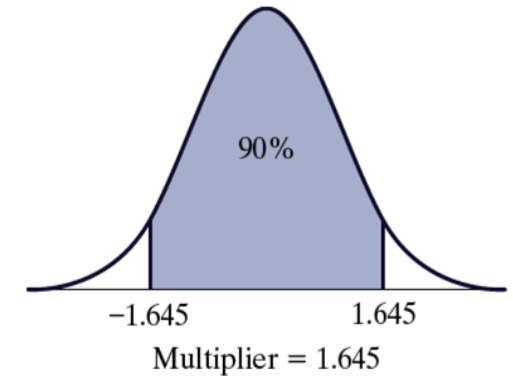


Technical Conditions for Validity of This Procedure

- Simple random sample from population of interest
 - $n \times \hat{p} \geq 10$ and $n \times (1 - \hat{p}) \geq 10$
- The *midpoint* of this interval is the sample proportion $\hat{p}$.
- The *width* of this interval is the difference in the endpoints.
 - Margin-of-error is the half-width of the interval.
- The margin-of-error depends on three factors:
 - confidence level, which determines z^*
 - sample size n
 - sample proportion $\hat{p}$ (to a lesser extent)
- Values of z^* for some common confidence levels are:

Confidence level	80%	90%	95%	99%	99.9%
Critical value z^*	1.282	1.645	1.960	2.576	3.291

- Applet: [Theory-Based Inference](#)



Confidence Interval Statement

- **Incorrect Statement:** 42.0% of all U.S. adults thought the economy was improving.
- **Correct Statement:** “We are 95% confident that between 40.4% and 43.6% of U.S. adults thought the economy was improving.”
 - Statements like this are called confidence intervals. They don’t tell us everything we might want to know, but they’re the best we can do.

Confidence Interval Simulation



Reese's Pieces® candies come in three colors: orange, yellow, and brown. Suppose that you take a random sample of candies and want to estimate the long-run proportion of candies that is orange.

Let's assume for now (although we would not know this when conducting the study) that this long-run proportion, symbolized by π , is equal to 0.45.

Reese's Pieces - 3 Questions



Suppose that you take **a random sample** of 100 Reese's Pieces candies.

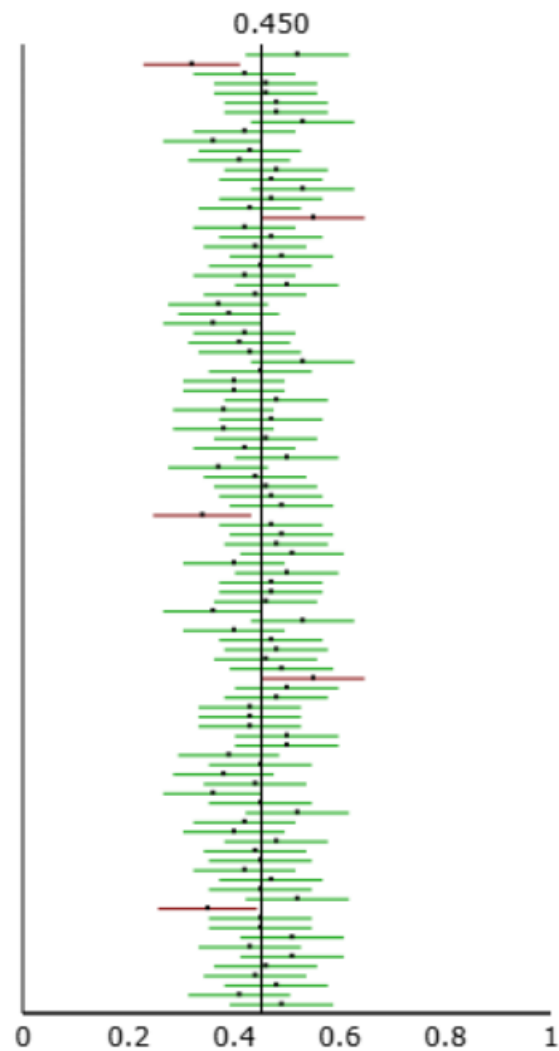
1. Is there any guarantee that the sample proportion will equal 0.45?
2. Is there any guarantee that its CI interval will contain the value 0.45?

Suppose that you select **another random sample** of 100 Reese's Pieces candies.

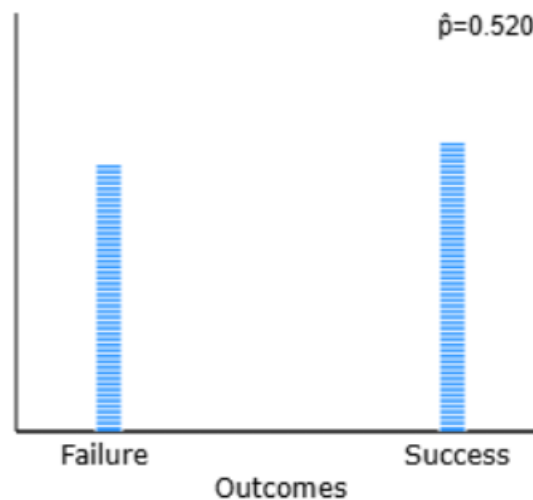
1. Is there any guarantee that the **sample proportion & CI** will be the same as for the first sample? How do you think they will differ?

We will use our [applet](#) for Simulating Confidence Intervals to understand what it means to say that we have “95% confidence” that an interval contains the parameter of interest?

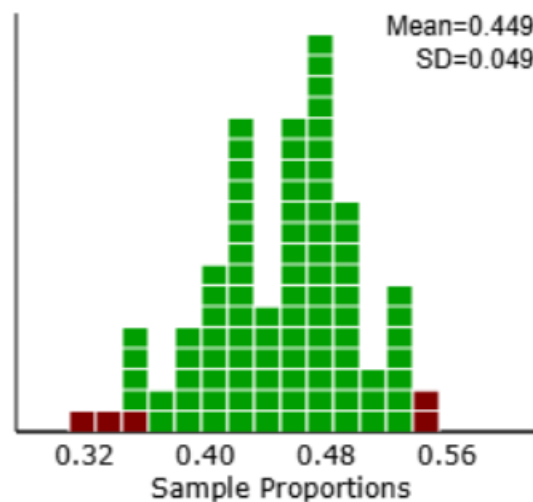
Confidence intervals



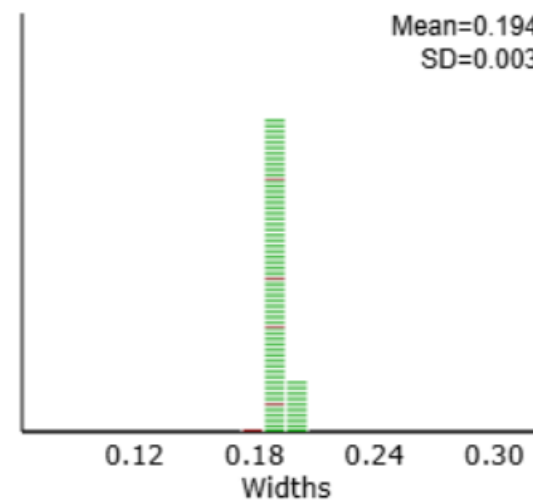
Most Recent Sample



Sample statistics (CI midpoints)



(CI widths)



Theoretical Sampling Distribution

